scenario1_analysis

May 25, 2026

1 Scenario 1: Operational Baseline Analysis

This notebook reviews the baseline capture for the smart home setup. It filters the Zeek logs, focuses on the target devices, and summarizes the main traffic patterns.

Topology: * Camera: `10.0.0.1` * Thermostat: `10.0.0.2` * Attacker: `10.0.0.3` * Gateway: `10.0.0.254` * Duration: 20 minutes

[1]:

```python
import os
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.ticker as mticker
import seaborn as sns
from zat.log_to_dataframe import LogToDataFrame

# Set academic plotting style for LaTeX embedding
sns.set_theme(style="whitegrid")
plt.rcParams.update({
    "font.family": "serif",
    "axes.labelsize": 12,
    "font.size": 12,
    "legend.fontsize": 10,
    "xtick.labelsize": 10,
    "ytick.labelsize": 10,
    "figure.dpi": 300
})

# Define paths
LOG_DIR = "../../zeek_logs/scenario1_logs/"  # Adjust relative path if needed
OUTPUT_DIR = "../../figures/scenario1/"
os.makedirs(OUTPUT_DIR, exist_ok=True)


def save_figure(fig, filename_base, output_dir=OUTPUT_DIR):
    pdf_path = os.path.join(output_dir, f"{filename_base}.pdf")
    png_path = os.path.join(output_dir, f"{filename_base}.png")
    fig.savefig(pdf_path, format="pdf", bbox_inches="tight")
```

```python
    fig.savefig(png_path, format="png", bbox_inches="tight", dpi=300)
    print(f"Saved figure: {pdf_path} and {png_path}")


def format_elapsed_mmss(value, _):
    total_seconds = max(0, int(round(value * 60)))
    minutes, seconds = divmod(total_seconds, 60)
    return f"{minutes:02d}:{seconds:02d}"
```

1.1 Data Ingestion & Preprocessing

We load the Zeek logs, clean the data, and keep only traffic from the target devices and gateway.

[2]:
```python
# Initialize ZAT parser
log_to_df = LogToDataFrame()

# Ingest Zeek Logs
conn_df = log_to_df.create_dataframe(os.path.join(LOG_DIR, 'conn.log'))
dns_df = log_to_df.create_dataframe(os.path.join(LOG_DIR, 'dns.log'))
ntp_df = log_to_df.create_dataframe(os.path.join(LOG_DIR, 'ntp.log'))

# Deduplicate packets caused by Mininet capturing on 'any' interface
conn_df = conn_df.drop_duplicates(subset=['uid'], keep='first')
dns_df = dns_df.drop_duplicates(subset=['uid'], keep='first')
ntp_df = ntp_df.drop_duplicates(subset=['uid'], keep='first')

# Preprocessing: Ensure timestamps are timezone-naive for easier math
if conn_df.index.tz is not None:
    conn_df.index = conn_df.index.tz_localize(None)
if dns_df.index.tz is not None:
    dns_df.index = dns_df.index.tz_localize(None)
if ntp_df.index.tz is not None:
    ntp_df.index = ntp_df.index.tz_localize(None)

# Define our forensic scope
target_ips = ['10.0.0.1', '10.0.0.2', '10.0.0.3', '10.0.0.254']

# Filter logs to ensure we capture any conversation touching our target␣␣
↪architecture
conn_df = conn_df[conn_df['id.orig_h'].isin(target_ips) | conn_df['id.resp_h'].
↪isin(target_ips)]
dns_df = dns_df[dns_df['id.orig_h'].isin(target_ips) | dns_df['id.resp_h'].
↪isin(target_ips)]
ntp_df = ntp_df[ntp_df['id.orig_h'].isin(target_ips) | ntp_df['id.resp_h'].
↪isin(target_ips)]
```

1.2 Macro Statistics and Baseline Check

This section compares expected traffic with the capture to make sure the baseline is clean.

- Baseline: no extra traffic.
- Later scenarios: unexpected traffic should stand out quickly.

[3]:
```python
# ==========================================
# MACRO STATS & MUD PROFILE VALIDATION
# ==========================================
import pandas as pd

# Get total connections tracked by Zeek
total_conns = len(conn_df)

# Calculate empirical capture duration
capture_duration = (conn_df.index.max() - conn_df.index.min()).total_seconds()

# Isolate traffic based on theoretical MUD Profiles
# Infrastructure Services
dns_conns = conn_df[(conn_df['id.resp_p'] == 53) & (conn_df['proto'] == 'udp')]
ntp_conns = conn_df[(conn_df['id.resp_p'] == 123) & (conn_df['proto'] == 'udp')]

# Smart Camera (10.0.0.1)
cam_udp_feed = conn_df[(conn_df['id.orig_h'] == '10.0.0.1') & (conn_df['id.
↪resp_p'] == 50005) & (conn_df['proto'] == 'udp')]
cam_tcp_control = conn_df[(conn_df['id.orig_h'] == '10.0.0.1') & (conn_df['id.
↪resp_p'] == 50005) & (conn_df['proto'] == 'tcp')]
cam_tls = conn_df[(conn_df['id.orig_h'] == '10.0.0.1') & (conn_df['id.resp_p']␣
↪== 443) & (conn_df['proto'] == 'tcp')]

# Smart Thermostat (10.0.0.2)
therm_mqtt = conn_df[(conn_df['id.orig_h'] == '10.0.0.2') & (conn_df['id.
↪resp_p'] == 8883) & (conn_df['proto'] == 'tcp')]
# Safely handle potential NaN values by filling them with 0 before evaluation
therm_keepalive = therm_mqtt[therm_mqtt['orig_bytes'].fillna(0) < 20]
therm_payload = therm_mqtt[therm_mqtt['orig_bytes'].fillna(0) >= 20]

# Treat expected background ICMP chatter as environmental noise
noise_conns = conn_df[
    (conn_df['proto'] == 'icmp') &
    (conn_df['id.orig_h'] == '10.0.0.254') &
    (conn_df['id.resp_h'] == '10.0.0.1')
]

# Sum all classified, authorized traffic
classified_sum = (
    len(dns_conns) + len(ntp_conns) +
```

```
    len(therm_keepalive) + len(therm_payload) +
    len(cam_udp_feed) + len(cam_tcp_control) + len(cam_tls)
)

# Print the Macro Statistics Report
print("=== MACRO NETWORK STATISTICS ===")
print(f"Total Connections Logged (conn_df): {total_conns}")
print(f"Empirical Capture Duration:         {capture_duration:.2f} seconds\n")

print("--- Classified Traffic Breakdown ---")
print(f"DNS Queries (UDP/53):               {len(dns_conns)}")
print(f"NTP Queries (UDP/123):              {len(ntp_conns)}")
print(f"Thermostat Keep-Alives (TCP/8883):  {len(therm_keepalive)}")
print(f"Thermostat Payloads (TCP/8883):     {len(therm_payload)}")
print(f"Camera Video Feed (UDP/50005):      {len(cam_udp_feed)}")
print(f"Camera TCP Control (TCP/50005):     {len(cam_tcp_control)}")
print(f"Camera TLS Telemetry (TCP/443):      {len(cam_tls)}\n")
print(f"Background ICMP Noise (Gateway->Camera): {len(noise_conns)}\n")

print("--- Connection Duration Analysis ---")
for idx, row in cam_tcp_control.iterrows():
    raw_duration = row['duration']

    # Check if duration is missing/NaN
    if pd.isna(raw_duration):
        flow_duration = 0.0
    # Check if it's a Pandas Timedelta object, if so extract total seconds
    elif isinstance(raw_duration, pd.Timedelta):
        flow_duration = raw_duration.total_seconds()
    # Fallback if it's already a float or integer
    else:
        flow_duration = float(raw_duration)

    print(f"Camera Control Channel | Timestamp: {idx.time()} | State:␣
 ↪{row['conn_state']} | Duration: {flow_duration:.2f}s")

# The Tripwire Validation
print("\n--- Baseline Validation ---")
print(f"Sum of Classified Authorized Connections: {classified_sum}")

unaccounted = total_conns - classified_sum
unaccounted = unaccounted - len(noise_conns)

if unaccounted == 0:
    print("SUCCESS: 100% of network traffic matches expected MUD profiles. Zero␣
 ↪unaccounted flows.")
else:
```

```
    print(f"FORENSIC ALERT: {unaccounted} unaccounted connections detected!")

    # Corrected exclusion logic including the camera control channel
    unaccounted_df = conn_df[
        ~conn_df.index.isin(dns_conns.index) &
        ~conn_df.index.isin(ntp_conns.index) &
        ~conn_df.index.isin(therm_keepalive.index) &
        ~conn_df.index.isin(therm_payload.index) &
        ~conn_df.index.isin(cam_udp_feed.index) &
        ~conn_df.index.isin(cam_tcp_control.index) &
        ~conn_df.index.isin(cam_tls.index) &
        ~conn_df.index.isin(noise_conns.index)
    ]
    display(unaccounted_df[['id.orig_h', 'id.resp_h', 'id.resp_p', 'proto',␣
  ↪'conn_state', 'duration']])
```

```
=== MACRO NETWORK STATISTICS ===
Total Connections Logged (conn_df): 181
Empirical Capture Duration:         1200.16 seconds

--- Classified Traffic Breakdown ---
DNS Queries (UDP/53):               42
NTP Queries (UDP/123):              6
Thermostat Keep-Alives (TCP/8883):  121
Thermostat Payloads (TCP/8883):     5
Camera Video Feed (UDP/50005):      1
Camera TCP Control (TCP/50005):     0
Camera TLS Telemetry (TCP/443):      5

Background ICMP Noise (Gateway->Camera): 1

--- Connection Duration Analysis ---

--- Baseline Validation ---
Sum of Classified Authorized Connections: 180
SUCCESS: 100% of network traffic matches expected MUD profiles. Zero unaccounted
flows.
```

1.3 Camera Behaviours

1.3.1 Camera Streaming

We measure the camera's UDP stream to see if it stays steady over time.

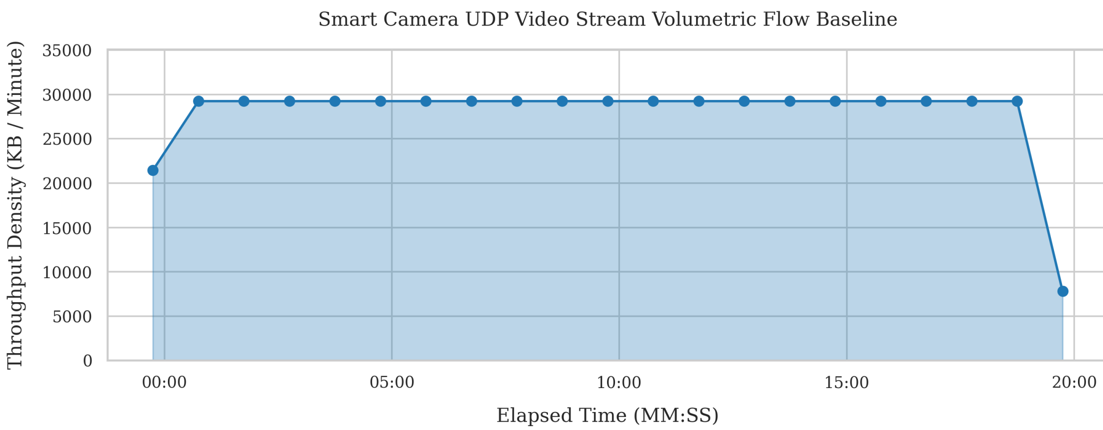
This gives a simple baseline for spotting unusual spikes later.

[4]:
```
# Isolate the Camera UDP Feed
camera_udp = conn_df[(conn_df['id.orig_h'] == '10.0.0.1') &
                     (conn_df['id.resp_p'] == 50005) &
```

```python
                    (conn_df['proto'] == 'udp')].copy()

# Fix Mininet 'any' interface double-counting for stateless UDP
camera_udp['orig_bytes'] = camera_udp['orig_bytes'] / 2

total_bytes = camera_udp['orig_bytes'].sum()
print(f"Camera Stream Total Originated Bytes: {total_bytes:,.0f} Bytes")

# ==========================================
# Universal Forensic Data Distribution
# ==========================================
# Get the global start and end of the entire capture
capture_start = conn_df.index.min()
capture_end = conn_df.index.max()

# Create a blank, continuous 1-second timeline for the whole capture
timeline = pd.Series(0.0, index=pd.date_range(start=capture_start,␣
 ↪end=capture_end, freq='1s'))

# Dynamically distribute the bytes for EVERY row Zeek logged
for flow_start, row in camera_udp.iterrows():
    raw_duration = row['duration']

    # Safely extract duration handling NaNs and Timedeltas
    if pd.isna(raw_duration):
        duration_sec = 1.0
    elif isinstance(raw_duration, pd.Timedelta):
        duration_sec = raw_duration.total_seconds()
    else:
        duration_sec = float(raw_duration)

    if duration_sec <= 0: duration_sec = 1.0 # Prevent division by zero

    # Calculate bytes per second for this specific Zeek row
    bytes_per_sec = row['orig_bytes'] / duration_sec
    flow_end = flow_start + pd.Timedelta(seconds=duration_sec)

    # Slice the timeline and inject the bandwidth
    active_window = (timeline.index >= flow_start) & (timeline.index < flow_end)
    timeline.loc[active_window] += bytes_per_sec

# Resample our distributed 1-second timeline into 1-minute bins
volume_over_time = timeline.resample('1min').sum()


def parse_kbit_rate(rate: str) -> int:
    value = rate.strip().lower()
```

```python
    if value.endswith('k'):
        return int(float(value[:-1]) * 1000)
    if value.endswith('m'):
        return int(float(value[:-1]) * 1000 * 1000)
    return int(float(value))

def expected_kb_per_min(rate: str) -> float:
    rate_bps = parse_kbit_rate(rate)
    return (rate_bps / 8.0) / 1024.0 * 60.0

# ==========================================
# Plotting (With Elapsed Time Fix)
# ==========================================
fig, ax = plt.subplots(figsize=(10, 4))
kb_values = volume_over_time.values / 1024

# Shift the timestamps so the graph starts at elapsed time 00:00
elapsed_time_index = volume_over_time.index - capture_start + pd.to_datetime('1970-01-01')

# Plot the area chart
ax.fill_between(elapsed_time_index, kb_values, color='#1f77b4', alpha=0.3)
ax.plot(elapsed_time_index, kb_values, marker='o', linestyle='-', color='#1f77b4', linewidth=1.5)

ax.set_title("Smart Camera UDP Video Stream Volumetric Flow Baseline", pad=15)
ax.set_xlabel("Elapsed Time (MM:SS)", labelpad=10)
ax.set_ylabel("Throughput Density (KB / Minute)", labelpad=10)

# Format the X-axis for elapsed minutes and seconds
ax.xaxis.set_major_formatter(plt.matplotlib.dates.DateFormatter('%M:%S'))

# Give the chart breathing room on top, handle edge cases if data is 0
top_limit = max(kb_values) * 1.2 if max(kb_values) > 0 else 1000
ax.set_ylim(bottom=0, top=top_limit)

plt.tight_layout()
save_figure(fig, "camera_volumetric_flow")

plt.show()

# Print empirical average to terminal
avg_kb_min = kb_values[kb_values > 0].mean() if len(kb_values[kb_values > 0]) > 0 else 0
target_kb_min = expected_kb_per_min("4000K")
print(f"Empirical Average Bandwidth: {avg_kb_min:,.2f} KB/minute")
print(f"Target Bandwidth (4000K):       {target_kb_min:,.2f} KB/minute")
```

```
Camera Stream Total Originated Bytes: 599,068,800 Bytes
Saved figure: ../../figures/scenario1/camera_volumetric_flow.pdf and
../../figures/scenario1/camera_volumetric_flow.png
```




```
Empirical Average Bandwidth: 27,858.46 KB/minute
Target Bandwidth (4000K):       29,296.88 KB/minute
```

1.3.2 Camera TLS Telemetry and Network Services

We check how often the camera phones home and how often the support services run.

- Camera TLS: 5 check-ins.
- NTP: 6 syncs.
- DNS: 42 lookups.

[5]:
```
# ==========================================
# INFRASTRUCTURE & BACKGROUND SERVICES VALIDATION
# ==========================================

# Camera TLS Telemetry Verification
camera_tls = conn_df[
    (conn_df['id.orig_h'] == '10.0.0.1') &
    (conn_df['id.resp_p'] == 443) &
    (conn_df['proto'] == 'tcp')
]
assert len(camera_tls) == 5, f"CRITICAL baseline failure: Expected 5 TLS flows,␣
↪found {len(camera_tls)}"
print("Camera TLS Telemetry Validated: Exactly 5 Flows detected (300s interval␣
↪heartbeat, inclusive of t=0).")

# Network Services (NTP) Verification
ntp_flows = conn_df[
    (conn_df['id.orig_h'].isin(['10.0.0.1', '10.0.0.2'])) &
```

```
    (conn_df['id.resp_p'] == 123) &
    (conn_df['proto'] == 'udp')
]
assert len(ntp_flows) == 6, f"CRITICAL baseline failure: Expected 6 NTP flows,␣
 ↪found {len(ntp_flows)}"
camera_ntp = len(ntp_flows[ntp_flows['id.orig_h'] == '10.0.0.1'])
therm_ntp = len(ntp_flows[ntp_flows['id.orig_h'] == '10.0.0.2'])
print(f"NTP Synchronization Validated: Total 6 Flows ({camera_ntp} Camera,␣
 ↪{therm_ntp} Thermostat).")

# Domain Name Resolution (DNS) Verification
dns_queries = dns_df[dns_df['id.orig_h'].isin(['10.0.0.1', '10.0.0.2'])]
assert len(dns_queries) == 42, f"CRITICAL baseline failure: Expected 42 DNS␣
 ↪queries, found {len(dns_queries)}"
camera_dns = len(dns_queries[dns_queries['id.orig_h'] == '10.0.0.1'])
therm_dns = len(dns_queries[dns_queries['id.orig_h'] == '10.0.0.2'])
print(f"DNS Queries Validated: Total 42 Queries ({camera_dns} Camera,␣
 ↪{therm_dns} Thermostat).")
```

```
Camera TLS Telemetry Validated: Exactly 5 Flows detected (300s interval
heartbeat, inclusive of t=0).
NTP Synchronization Validated: Total 6 Flows (3 Camera, 3 Thermostat).
DNS Queries Validated: Total 42 Queries (21 Camera, 21 Thermostat).
```

1.3.3 Camera Traffic Balance

We compare the camera's outgoing bytes with the bytes it receives.

Normal traffic should be mostly outgoing. If incoming traffic rises a lot, that can signal unusual control or remote access.

This check still works even if the capture duplicates some bytes.

[6]:
```
# ==========================================
# Camera Directional Byte Ratio Analysis
# ==========================================

# Isolate all connections initiated by the Camera
camera_flows = conn_df[conn_df['id.orig_h'] == '10.0.0.1'].copy()

# Sum the Outbound (orig) and Inbound (resp) bytes securely
outbound_bytes = camera_flows['orig_bytes'].fillna(0).sum()
inbound_bytes = camera_flows['resp_bytes'].fillna(0).sum()
total_bytes = outbound_bytes + inbound_bytes

# Calculate the percentage distribution
if total_bytes > 0:
    outbound_ratio = (outbound_bytes / total_bytes) * 100
```

```
    inbound_ratio = (inbound_bytes / total_bytes) * 100
else:
    outbound_ratio = 0
    inbound_ratio = 0

# Professional Output Report
print("=== CAMERA DIRECTIONAL BYTE RATIO ===")
print(f"Total Traffic Volume: {total_bytes:,.0f} Bytes\n")

print(f"Outbound Transmitted (orig_bytes): {outbound_bytes:,.0f} Bytes␣
 ↪({outbound_ratio:.4f}%)")
print(f"Inbound Received (resp_bytes):     {inbound_bytes:,.0f} Bytes␣
 ↪({inbound_ratio:.4f}%)\n")

# Automated Tripwire
if outbound_ratio > 99.0:
    print("Baseline Validated: Traffic is highly asymmetric (>99% outbound),␣
 ↪proving normal transmitter behavior.")
else:
    print("FORENSIC ALERT: High inbound volume detected! Potential Reverse␣
 ↪Shell or C2 interaction.")
```

```
=== CAMERA DIRECTIONAL BYTE RATIO ===
Total Traffic Volume: 1,198,143,151 Bytes

Outbound Transmitted (orig_bytes): 1,198,140,463 Bytes (99.9998%)
Inbound Received (resp_bytes):     2,688 Bytes (0.0002%)

Baseline Validated: Traffic is highly asymmetric (>99% outbound), proving normal
transmitter behavior.
```

1.4 Thermostat Behavior

1.4.1 Keep-Alives and Telemetry

We separate the thermostat's small keep-alive messages from its larger sensor updates and check that the pattern stays regular.

If that pattern changes, it can point to a problem.

[7]:
```
# Isolate for Thermostat MQTT traffic (Enforcing TCP protocol)
thermostat_mqtt = conn_df[
    (conn_df['id.orig_h'] == '10.0.0.2') &
    (conn_df['id.resp_p'] == 8883) &
    (conn_df['proto'] == 'tcp')
].copy()

# Sort by timestamp (index) to accurately calculate consecutive timing
```

```python
thermostat_mqtt = thermostat_mqtt.sort_index()

# Separate Keep-Alives from Telemetry using a 20-byte threshold
# Handle NaN values in orig_bytes to prevent accidental row dropping
keep_alives = thermostat_mqtt[thermostat_mqtt['orig_bytes'].fillna(0) < 20].
↪copy()
telemetry = thermostat_mqtt[thermostat_mqtt['orig_bytes'].fillna(0) >= 20].
↪copy()

# ---------------------------------------------------------
# Dynamic Ground Truth Validation (The Fencepost Math)
# ---------------------------------------------------------
# Calculate the exact span of the MQTT capture in seconds
actual_duration = (thermostat_mqtt.index.max() - thermostat_mqtt.index.min()).
↪total_seconds()

keep_alive_interval = 10.0
telemetry_interval = 300.0

# The +1 accounts for the initial packet sent at t=0
expected_keep_alives = int(actual_duration / keep_alive_interval) + 1
expected_telemetry = int(actual_duration / telemetry_interval) + 1

# Assertions with a +/- 2 packet tolerance for virtualization boundary timing
assert abs(len(keep_alives) - expected_keep_alives) <= 2, f"Expected␣
↪~{expected_keep_alives} keep-alives, found {len(keep_alives)}"
assert abs(len(telemetry) - expected_telemetry) <= 2, f"Expected␣
↪~{expected_telemetry} telemetry flows, found {len(telemetry)}"

print("=== MQTT BEHAVIORAL VALIDATION ===")
print(f"Empirical MQTT Capture Span: {actual_duration:.2f} seconds\n")
print(f"Keep-Alives Validated: {len(keep_alives)} Flows (Target:␣
↪~{expected_keep_alives})")
print(f"Telemetry Validated:   {len(telemetry)} Flows (Target:␣
↪~{expected_telemetry})\n")

# ---------------------------------------------------------
# Inter-Arrival Time (IAT) Calculation
# ---------------------------------------------------------
# Calculate time difference between consecutive keep-alive rows
keep_alives['IAT'] = keep_alives.index.to_series().diff().dt.total_seconds()

# CRITICAL FIX: Drop the first row's NaN IAT before doing statistical math
iat_clean = keep_alives['IAT'].dropna()

iat_mean = iat_clean.mean()
```

```
iat_var = iat_clean.var()

print("--- Temporal Fingerprint (Keep-Alives) ---")
print(f"IAT Mean:     {iat_mean:.4f} seconds")
print(f"IAT Variance: {iat_var:.6f}")
```

```
=== MQTT BEHAVIORAL VALIDATION ===
Empirical MQTT Capture Span: 1200.10 seconds

Keep-Alives Validated: 121 Flows (Target: ~121)
Telemetry Validated:   5 Flows (Target: ~5)

--- Temporal Fingerprint (Keep-Alives) ---
IAT Mean:     10.0001 seconds
IAT Variance: 0.000449
```

1.4.2 Keep-Alive Timing

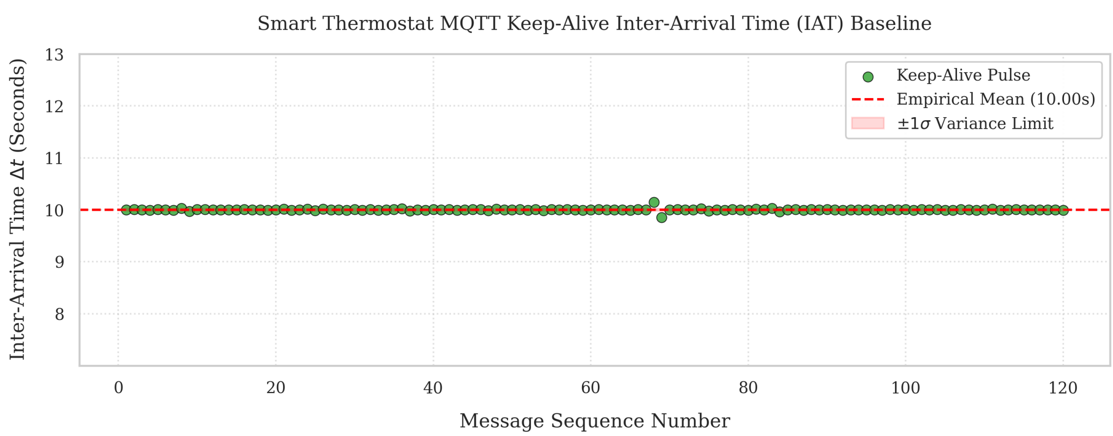
We plot the time between thermostat keep-alives to see if they arrive on a steady schedule.

A stable baseline should look tight and regular.

[8]:
```
# Clean the NaN value from the first row (the initial fencepost packet)
plot_data = keep_alives.dropna(subset=['IAT']).copy()

# Re-sequence from 1 to N after dropping the first row
plot_data['Sequence'] = range(1, len(plot_data) + 1)

# Calculate plotting statistics
iat_mean = plot_data['IAT'].mean()
iat_std = plot_data['IAT'].std()

# Academic Plotting
fig, ax = plt.subplots(figsize=(10, 4))

# Plot the individual IAT events
ax.scatter(plot_data['Sequence'], plot_data['IAT'],
           color='#2ca02c', alpha=0.8, edgecolors='black', linewidth=0.5, s=40,␣
 ↪label='Keep-Alive Pulse')

# Plot the empirical mean
ax.axhline(y=iat_mean, color='red', linestyle='--', linewidth=1.5,␣
 ↪label=f"Empirical Mean ({iat_mean:.2f}s)")

# Add a shaded band for +/- 1 Standard Deviation to visually prove the␣
 ↪near-zero variance
ax.fill_between(plot_data['Sequence'],
                iat_mean - iat_std,
```

```
                iat_mean + iat_std,
                color='red', alpha=0.15, label=r'$\pm 1\sigma$ Variance Limit')

ax.set_title("Smart Thermostat MQTT Keep-Alive Inter-Arrival Time (IAT)␣
  ↪Baseline", pad=15)
ax.set_xlabel("Message Sequence Number", labelpad=10)
ax.set_ylabel("Inter-Arrival Time $\Delta t$ (Seconds)", labelpad=10)

# Add a subtle grid to make temporal deviations easy to spot
ax.grid(True, linestyle=':', alpha=0.6)

# Focus the Y-axis tightly around the expected baseline to highlight stability
ax.set_ylim(bottom=max(0, iat_mean - 3), top=iat_mean + 3)

# Place the legend neatly in the corner
ax.legend(loc="upper right", framealpha=0.95)

plt.tight_layout()
save_figure(fig, "thermostat_iat_distribution")

plt.show()

print(f"IAT Scatter Plot Generated.")
print(f"   - Mapped {len(plot_data)} sequential pulses.")
print(f"   - Empirical Mean: {iat_mean:.4f}s | Standard Deviation: {iat_std:.
  ↪4f}s")
```

```
Saved figure: ../../figures/scenario1/thermostat_iat_distribution.pdf and
../../figures/scenario1/thermostat_iat_distribution.png
```




```
IAT Scatter Plot Generated.
   - Mapped 120 sequential pulses.
```

```
- Empirical Mean: 10.0001s | Standard Deviation: 0.0212s
```

1.4.3 Telemetry Size

We measure how large the thermostat's sensor messages are.

They should stay close in size, and a big change could mean something is off.

[9]:
```python
# ==========================================
# Thermostat Telemetry Payload Footprinting
# ==========================================

# Extract the byte sizes of the telemetry packets (>= 20 bytes)
# We use the 'telemetry' dataframe previously generated in the IAT cell
payload_sizes = telemetry['orig_bytes'].dropna()

# Calculate the statistical footprint
payload_mean = payload_sizes.mean()

# Variance requires at least 2 packets. If identical, variance is␣
↪mathematically 0.0
payload_var = payload_sizes.var() if len(payload_sizes) > 1 else 0.0

# Min/Max boundary detection
payload_min = payload_sizes.min()
payload_max = payload_sizes.max()

# Professional Output Report
print("=== MQTT TELEMETRY PAYLOAD FOOTPRINT ===")
print(f"Total Telemetry Reports Analyzed: {len(payload_sizes)}\n")

print(f"Empirical Mean Size: {payload_mean:.2f} Bytes")
print(f"Payload Variance:    {payload_var:.4f}")
print(f"Observed Range:      {payload_min:.0f} Bytes to {payload_max:.0f}␣
↪Bytes\n")

# Automated Tripwire
# A variance of less than 5 indicates extreme structural rigidity (expected for␣
↪M2M)
if payload_var < 5.0:
    print("Baseline Validated: Telemetry payloads are structurally rigid and␣
↪predictable.")
else:
    print("FORENSIC ALERT: High payload variance detected! Potential data␣
↪exfiltration or payload stuffing.")
```

```
=== MQTT TELEMETRY PAYLOAD FOOTPRINT ===
Total Telemetry Reports Analyzed: 5
```

```
Empirical Mean Size: 27.00 Bytes
Payload Variance:    0.0000
Observed Range:      27 Bytes to 27 Bytes

Baseline Validated: Telemetry payloads are structurally rigid and predictable.
```

1.5 Network Health

1.5.1 TCP Connection States

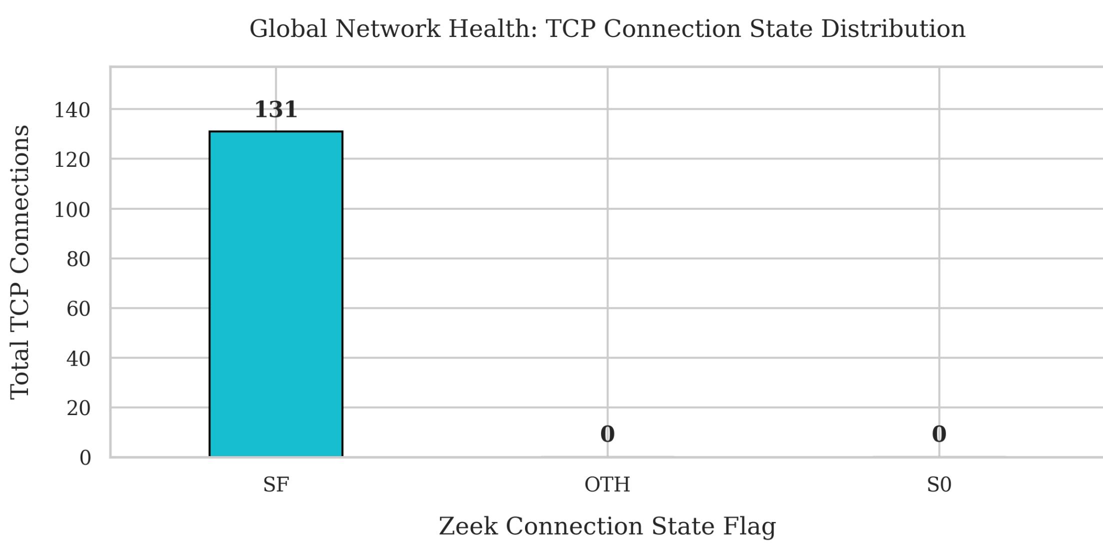
We look at the main TCP states to see whether the network is behaving normally.

Clean traffic should mostly show normal connections.

[10]:
```python
# ============================================
# 5.1 Global Network Health (TCP States)
# ============================================

# Isolate ONLY TCP traffic
tcp_df = conn_df[conn_df['proto'] == 'tcp'].copy()

# Compute state distribution
state_counts = tcp_df['conn_state'].value_counts()

# Academic Plotting
fig, ax = plt.subplots(figsize=(8, 4)) # Slightly wider to accommodate labels

# Use a distinct color palette
state_counts.plot(kind='bar', color='#17becf', ax=ax, width=0.4,␣
 ↪edgecolor='black')

ax.set_title("Global Network Health: TCP Connection State Distribution", pad=15)
ax.set_xlabel("Zeek Connection State Flag", labelpad=10)
ax.set_ylabel("Total TCP Connections", labelpad=10)

# Explicitly set the x-axis limits to give a single bar breathing room
ax.set_xlim(-0.5, len(state_counts) - 0.5)
plt.xticks(rotation=0)

# Bulletproof text annotation layer
for p in ax.patches:
    height = p.get_height()
    ax.annotate(f'{int(height)}',
                xy=(p.get_x() + p.get_width() / 2, height),
                xytext=(0, 5),  # 5 points vertical offset
                textcoords="offset points",
                ha='center', va='bottom', fontsize=11, weight='bold')
```

```python
# Ensure the Y-axis has room for the top annotation
ax.set_ylim(0, state_counts.max() * 1.2)

plt.tight_layout()
save_figure(fig, "tcp_connection_states")
plt.show()

# ---------------------------------------------------------
# Automated Contextual Reporting
# ---------------------------------------------------------
# A reference dictionary mapping Zeek states to their physical network reality
zeek_state_definitions = {
    'SF': 'Normal establishment and termination (Clean Teardown).',
    'S0': 'Connection attempt seen, no reply (Scanner / Blocked).',
    'S1': 'Connection established, not terminated (Persistent Socket).',
    'REJ': 'Connection attempt rejected by responder (RST sent).',
    'RSTO': 'Connection established, originator aborted (RST sent).',
    'RSTR': 'Established, responder aborted (RST sent).',
    'OTH': 'No SYN seen, just mid-stream traffic (Asymmetric routing).',
}

print("=== TCP STATE FORENSIC REPORT ===")
print(f"Total TCP Flows Analyzed: {len(tcp_df)}\n")

for state, count in state_counts.items():
    definition = zeek_state_definitions.get(state, "Unknown state.")
    print(f"[{state}] : {count:,.0f} Flows -> {definition}")
```

```
Saved figure: ../../figures/scenario1/tcp_connection_states.pdf and
../../figures/scenario1/tcp_connection_states.png
```



```
=== TCP STATE FORENSIC REPORT ===
Total TCP Flows Analyzed: 131

[SF] : 131 Flows -> Normal establishment and termination (Clean Teardown).
[OTH] : 0 Flows -> No SYN seen, just mid-stream traffic (Asymmetric routing).
[S0] : 0 Flows -> Connection attempt seen, no reply (Scanner / Blocked).
```

1.5.2 Flow Rate

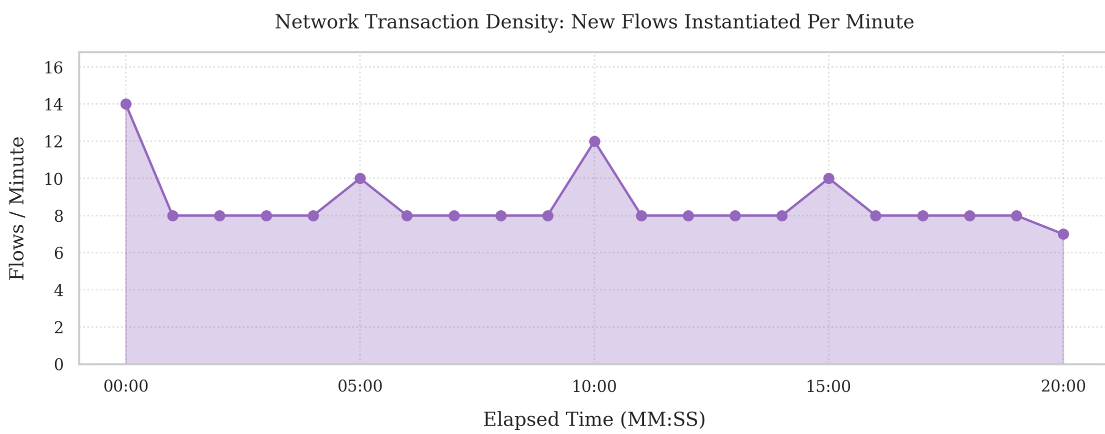
We count how many new connections appear each minute.

A steady baseline should stay low, while unusual activity should create spikes.

[11]:
```
# ==========================================
# 5.2 Network Density (Flows / Minute)
# ==========================================

# Resample to count total flows per minute.
# origin='start' ensures bins are exactly 60 seconds long starting from the␣
↪very first packet.
flows_per_minute = conn_df['uid'].resample('1min', origin='start').count()

# Apply the '1970' Elapsed Time trick to cleanly format the X-axis
capture_start = conn_df.index.min()
elapsed_time_index = flows_per_minute.index - capture_start + pd.
↪to_datetime('1970-01-01')

# Academic Plotting
fig, ax = plt.subplots(figsize=(10, 4))

# Using a fill_between area chart visually aligns with your Camera Volumetric␣
↪graph
ax.fill_between(elapsed_time_index, flows_per_minute.values, color='#9467bd',␣
↪alpha=0.3)
ax.plot(elapsed_time_index, flows_per_minute.values, color='#9467bd',␣
↪marker='o', linestyle='-', linewidth=1.5)

ax.set_title("Network Transaction Density: New Flows Instantiated Per Minute",␣
↪pad=15)
ax.set_xlabel("Elapsed Time (MM:SS)", labelpad=10)
ax.set_ylabel("Flows / Minute", labelpad=10)

# Standard matplotlib formatter (no custom functions required)
ax.xaxis.set_major_formatter(plt.matplotlib.dates.DateFormatter('%M:%S'))

# Add a subtle grid to help the reader track peaks
```

```
ax.grid(True, linestyle=':', alpha=0.6)

# Give the chart breathing room on top
top_limit = flows_per_minute.max() * 1.2 if flows_per_minute.max() > 0 else 10
ax.set_ylim(bottom=0, top=top_limit)

plt.tight_layout()
save_figure(fig, "network_flow_density")
plt.show()

# Contextual Output Report
print("=== NETWORK DENSITY REPORT ===")
print(f"Average Flows per Minute:      {flows_per_minute.mean():.2f}")
print(f"Peak Flows in a Single Minute: {flows_per_minute.max():.0f}")
```

```
Saved figure: ../../figures/scenario1/network_flow_density.pdf and
../../figures/scenario1/network_flow_density.png
```




```
=== NETWORK DENSITY REPORT ===
Average Flows per Minute:      8.62
Peak Flows in a Single Minute: 14
```

1.5.3 Traffic Map

We map which devices talk to each other.

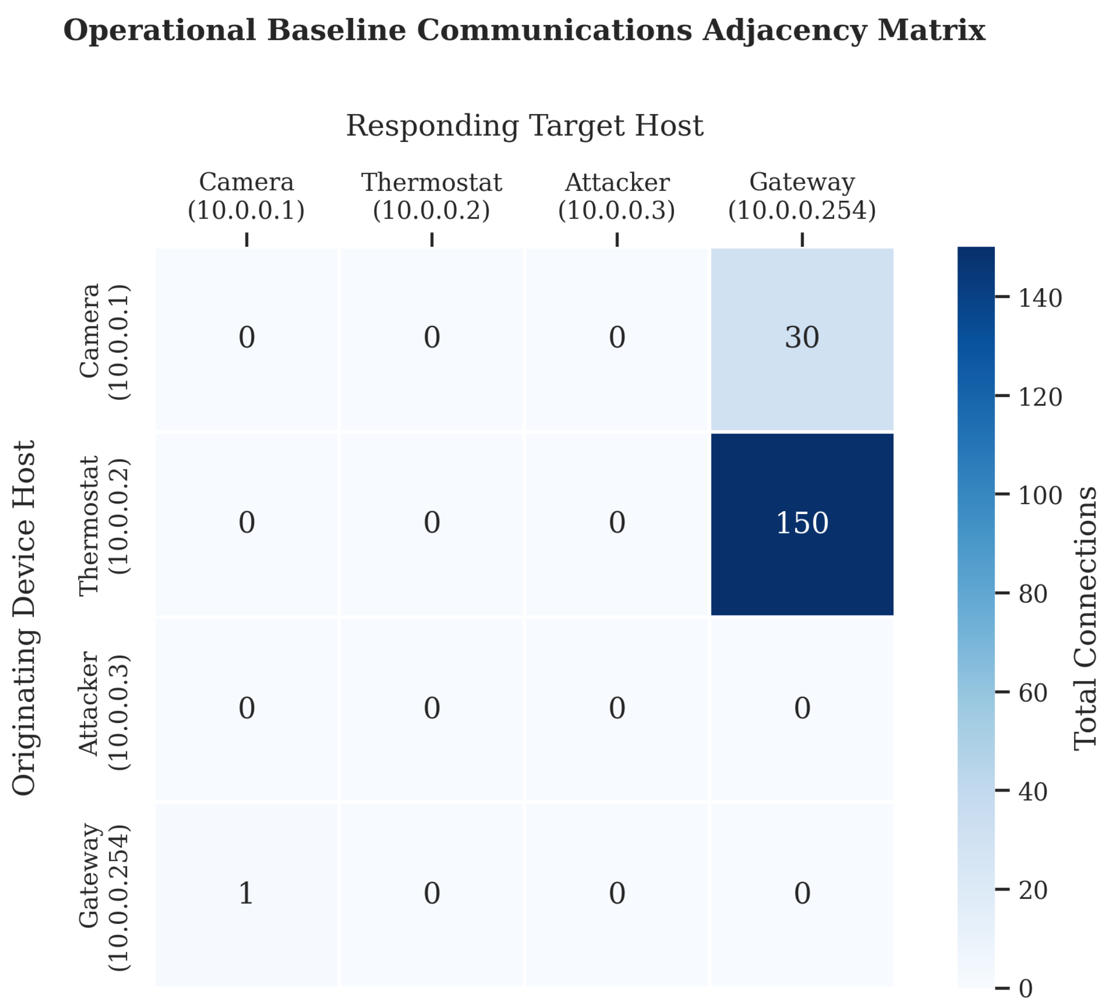
In the baseline, the camera and thermostat should only talk to the gateway, not to each other.

[12]:
```
# ==========================================
# Topology Matrix (Adjacency Heatmap)
# ==========================================

# Build the raw communication matrix
```

```python
matrix_data = pd.crosstab(conn_df['id.orig_h'], conn_df['id.resp_h'])

# Ensure all 4 core nodes are represented to protect baseline symmetry
# (target_ips was defined in Cell 1: ['10.0.0.1', '10.0.0.2', '10.0.0.3', '10.0.0.254'])
for ip in target_ips:
    if ip not in matrix_data.index:
        matrix_data.loc[ip] = 0
    if ip not in matrix_data.columns:
        matrix_data[ip] = 0

# Sort logically to build the matrix flow map
matrix_data = matrix_data.reindex(index=target_ips, columns=target_ips).fillna(0)

# CRITICAL UPGRADE: Human-Readable Device Mapping
device_labels = {
    '10.0.0.1': 'Camera\n(10.0.0.1)',
    '10.0.0.2': 'Thermostat\n(10.0.0.2)',
    '10.0.0.3': 'Attacker\n(10.0.0.3)',
    '10.0.0.254': 'Gateway\n(10.0.0.254)'
}

# Rename the axes for the heatmap
matrix_data_labeled = matrix_data.rename(index=device_labels, columns=device_labels)

# Academic Plotting
fig, ax = plt.subplots(figsize=(8, 6))

# Using a custom color map: 'Blues' is great, but we can make it starker for zeros
sns.heatmap(matrix_data_labeled, annot=True, fmt=".0f", cmap="Blues", cbar=True, square=True,
            linewidths=1, linecolor='white', cbar_kws={'label': 'Total Connections'}, ax=ax)

# Move the X-axis labels to the top for a more traditional matrix feel (Optional but looks great)
ax.xaxis.tick_top()
ax.xaxis.set_label_position('top')

ax.set_title("Operational Baseline Communications Adjacency Matrix", pad=30, weight='bold')
ax.set_xlabel("Responding Target Host", labelpad=15)
ax.set_ylabel("Originating Device Host", labelpad=15)
```

```python
plt.tight_layout()
save_figure(fig, "topology_traffic_matrix")
plt.show()

# ---------------------------------------------------------
# Automated Lateral Movement Validation
# ---------------------------------------------------------
print("=== TOPOLOGY RULES VALIDATION ===")

# Check if Camera (1) talked to Thermostat (2), or vice versa
cam_to_therm = matrix_data.loc['10.0.0.1', '10.0.0.2']
therm_to_cam = matrix_data.loc['10.0.0.2', '10.0.0.1']
lateral_traffic = cam_to_therm + therm_to_cam

if lateral_traffic == 0:
    print("Zero-Trust Baseline Validated: No lateral communication between IoT edge devices.")
else:
    print(f"FORENSIC ALERT: {lateral_traffic:,.0f} lateral connections detected between Camera and Thermostat!")
```

```
Saved figure: ../../figures/scenario1/topology_traffic_matrix.pdf and
../../figures/scenario1/topology_traffic_matrix.png
```




```
=== TOPOLOGY RULES VALIDATION ===
Zero-Trust Baseline Validated: No lateral communication between IoT edge
devices.
```

1.5.4 DNS Requests

We track DNS lookups over time to see when devices search for their services.

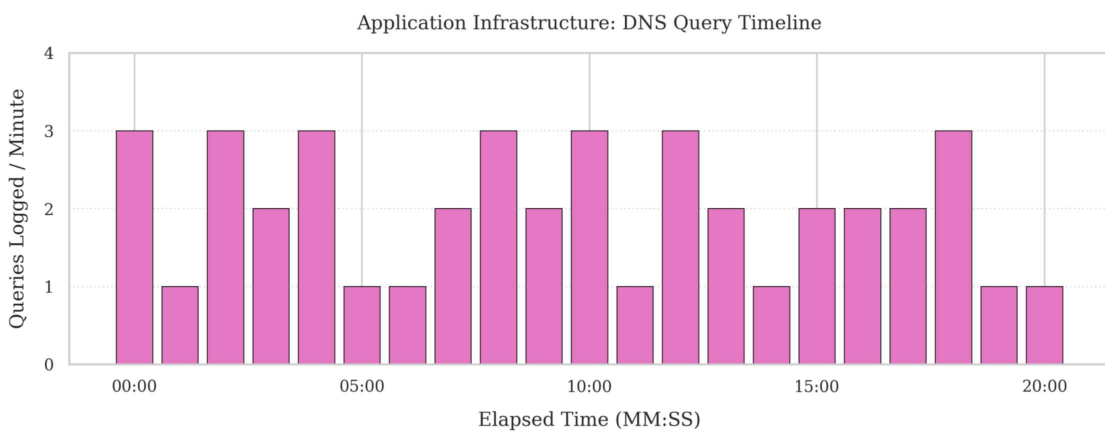
This also helps spot odd or unexpected lookups.

[13]:
```
# ==========================================
# Infrastructure Services (DNS Profiling)
# ==========================================

# Resample DNS query frequency
dns_frequency = dns_df['uid'].resample('1min', origin='start').count()
```

```python
# Apply the '1970' Elapsed Time trick
capture_start = conn_df.index.min()
elapsed_time_index = dns_frequency.index - capture_start + pd.
    ↪to_datetime('1970-01-01')

# Academic Plotting
fig, ax = plt.subplots(figsize=(10, 4))

# For a datetime x-axis, width is calculated in days.
# 1 minute = 1/(24*60) days. We multiply by 0.8 to leave gaps between bars.
bar_width = (1 / 1440) * 0.8

ax.bar(elapsed_time_index, dns_frequency.values, width=bar_width,␣
    ↪color='#e377c2', edgecolor='black', linewidth=0.5)

ax.set_title("Application Infrastructure: DNS Query Timeline", pad=15)
ax.set_xlabel("Elapsed Time (MM:SS)", labelpad=10)
ax.set_ylabel("Queries Logged / Minute", labelpad=10)

# Standard matplotlib formatter
ax.xaxis.set_major_formatter(plt.matplotlib.dates.DateFormatter('%M:%S'))

# Set y-axis to handle the low volume of DNS smoothly (forces integers on Y␣
    ↪axis)
ax.yaxis.get_major_locator().set_params(integer=True)
ax.set_ylim(bottom=0, top=max(3, dns_frequency.max() + 1))

# Add a subtle grid
ax.grid(True, axis='y', linestyle=':', alpha=0.6)

plt.tight_layout()
save_figure(fig, "dns_query_timeline")
plt.show()

# ---------------------------------------------------------
# Automated Domain Resolution Report
# ---------------------------------------------------------
print("=== DNS QUERY FORENSIC REPORT ===")
print(f"Total DNS Queries Logged: {len(dns_df)}\n")

# Check if the 'query' column exists (standard Zeek dns.log feature)
if 'query' in dns_df.columns:
    print("--- Queried Domains ---")
    domain_counts = dns_df['query'].value_counts()
    for domain, count in domain_counts.items():
        print(f"{domain} : {count} queries")
```

```python
    print("\n--- Baseline Validation ---")
    # Tripwire logic: If there are domains other than your 2 authorized ones, throw an alert
    unauthorized_domains = domain_counts[~domain_counts.index.isin(['api.smartcamera.com', 'mqtt.smartthermostat.com'])]

    if len(unauthorized_domains) == 0:
        print("Baseline Validated: 100% of DNS lookups match authorized IoT endpoints.")
    else:
        print(f"FORENSIC ALERT: {len(unauthorized_domains)} unauthorized domains requested! Potential DGA or DNS Tunneling.")
else:
    print(" Note: 'query' column not found in DataFrame. Ensure Zeek dns.log was parsed correctly.")
```

```
Saved figure: ../../figures/scenario1/dns_query_timeline.pdf and
../../figures/scenario1/dns_query_timeline.png
```




```
=== DNS QUERY FORENSIC REPORT ===
Total DNS Queries Logged: 42

--- Queried Domains ---
api.smartcamera.com : 21 queries
mqtt.smartthermostat.com : 21 queries

--- Baseline Validation ---
Baseline Validated: 100% of DNS lookups match authorized IoT endpoints.
```

1.5.5 Attacker Check

We search for any traffic involving the attacker IP.

In the baseline, there should be none.

[14]:
```python
# ==========================================
# Attacker Dormancy Validation
# ==========================================

attacker_ip = '10.0.0.3'

# Efficiently isolate any flow involving the attacker using the OR (|) operator
attacker_flows = conn_df[
    (conn_df['id.orig_h'] == attacker_ip) |
    (conn_df['id.resp_h'] == attacker_ip)
].copy()

total_attacker_flows = len(attacker_flows)

# Professional Output Report
print("=== ATTACKER DORMANCY REPORT ===")
print(f"Target Adversary Node: {attacker_ip}\n")

# Dynamic Tripwire and Evidence Display
if total_attacker_flows == 0:
    print(f"Baseline Scientifically Validated: 0 packets originating from or␣
↪destined to {attacker_ip}.")
    print("   The Scenario 1 control environment is pristine and ready for␣
↪comparison.")
else:
    print(f"FORENSIC ALERT: Baseline Contamination! Found␣
↪{total_attacker_flows} rogue flows involving the Attacker IP.")
    print("   Displaying evidence table for investigator review:")

    # Display the actual rogue packets so you can figure out what broke the␣
↪clean room
    display(attacker_flows[['id.orig_h', 'id.resp_p', 'id.resp_h', 'proto',␣
↪'conn_state', 'orig_bytes']])
```

```
=== ATTACKER DORMANCY REPORT ===
Target Adversary Node: 10.0.0.3

Baseline Scientifically Validated: 0 packets originating from or destined to
10.0.0.3.
   The Scenario 1 control environment is pristine and ready for comparison.
```